بسم الله الرحمن الرحيم

Basic Immunology

By

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Immunity:

- Is the **defensive mechanism** of the body against foreign substances entering it, e.g.:-
  - Bacteria,
  - Virus,
  - Microbial products.

- The goal is to **recognize, neutralize and destroy**.

- **Types of immunity**:

- Natural (innate or non specific).
- Acquired (specific).

- **Natural immunity**:

  It is effective but **not specific** and is not enhanced by previous exposure to foreign substances. (No memory).

- **1. Mechanical barrier and normal secretion:**

  Intact skin and mucous membrane afford a high degree of protection.

- ***A. Skin:***

- Indigestible **keratin** by microorganisms.
- High concentration of **salts in drying sweat**.
- **Bactericidal and fungicidal property** of sebaceous and sweat glands.

- ***B. Mucous membrane:***

1- Sticky mucous trap inhaled particles.

2- Cilia sweeps inhaled particles to oropharynx, then they are swallowed.

3- Mucopolysaccharides in nasal secretion and saliva block some viruses.

4- Acidic secretion of the stomach destroy most organisms.

5- Lysozymes (antibacterial) in secretion of respiratory tract, G.I.T. gut, saliva and tears.

6- Mechanical cleansing action of saliva, tears and flushing action of urine.

7- Peristalsis prevents bacterial overgrowth.

- 2. **Normal bacterial flora** e.g. symbiotic relationship between lactobacilli and yeast.
- 3. Naturally occurring **antimicrobial substances** e.g. interferon, lysozyme and complement.
- 4. **Cells** e.g. Natural Killer cells (NK) and phagocytic cells like neutrophils, macrophages.
- 5. **Body hormones** increase susceptibility to infection in **Diabetes, hyperthyroidism and adrenal dysfunction.**

- 6. **Genetic predisposition** certain species are susceptible to particular agents, while others are immunized to the same agent.

- 7. **Body temperature** pyrexia may function as a protective response against invading microorganisms.

- 8. **Oxygen tension** in the body may limit the growth of microorganisms.

- 9. **Inflammation, vasodilatation, chemotaxis and phagocytosis** all aim to get rid of organisms.

- 10.**Soluble proteins**: inflammatory processes as well as infection, burns and trauma induce changes of the host response termed **the acute phase response.**

- The response is associated with metabolic changes **in the liver in the form of hepatic synthesis** of certain proteins such as C-reactive protein, hepatoglobin, ceruloplasmin and complement components.

- **Soluble proteins:**
- 1. C-reactive protein (CRP)
- 2. α1-Antitraypsin (AAT)
- 3. Ceruloplasmin 4.Haptoglobin
- 5. Heat – shock protein
- 6. Lipopolysaccharide binding protein (LBP) and (CD14)
- 7. Mannose binding protein

- **Immunogen (antigen)**

  - Is a substance when introduced into the body is able to **produce specific immunological response** (production of antibodies) and is able to react with the products of that response.

- **Characters of the antigen (Immunogen):**

1. Foreignness to the animal.

2. High molecular weight:

- The higher the molecular weight, the more potent the antigen.

- The most potent antigens are those with a **molecular weight over 10.000 Dalton.**

* **Hapten** (incomplete or partial antigen)

- Antigen with a small molecular weight, alone has no antigenensity e.g. **most drugs**, but when introduced in the body it combines with body proteins forming hapten body-protein conjugate that provoke an immune response.

3. Chemical nature:

- The more the chemical complexity, the **more potent the antigen**, e.g. with synthetic peptides. Homopolymers consisted of repeated units of single amino-acid are poor immunogen while copolymers consisted of 2 or better 3 polypeptide chains are more active.

- **4. Antigenic determinant:**
- The antigen should have an antigenic determinant **"epitopes"** to react with the receptors of the lymphocytes.

- **Epitope can be defined as** that portion of an immunogen combining with the antibody.

- **5. Dose and route of administration:**
- In general, once the **threshold** is exceeded, increasing the dose increase the immune response. **S.C. & I.M. routes** of administration are the safest while I.V. is the most dangerous. .

- **6. Genetic constitution:**
- The ability of the body to respond to the antigen varies with the genetic make up (inherited as an autosomal dominant trait).

- **7. Degradability:**
- Immunogen must be a substance which can be **degraded by enzymes of antigen presenting cells.**

- **Super antigen**:

* Antigen eliciting a vigerous immune response

* **As an example**, Lipopolysaccharides of some bacteria and retroviruses

Acquired Immunity

- A specific immune mechanism.

- On subsequent exposure to foreign materials it becomes more efficient and effective in elimination of these foreign materials.

- **Components:**

1. Lymphocytes.
2. Phagocytic system.
3. Complement system.

Lymphocytes

1. **T-Lymphocytes:**
   Small, non granular concerned with cell mediated immunity.
   Represent 80-90% of circulating lymphocytes.

2. **B-Lymphocytes:**
   Small, non-granular concerned with humoral immunity.
   Represent 5-15% lymphocyte.

3. **Natural Killer cell N.K.**
   Large granular lymphocyte.
   Represent 5-10% of circulating lymphocytes.

Natural Killer Cell (NK)

- They are **large granular lymphocytes**, contain **azurophilic granules** in its cytoplasm representing 5-10% of lymphocytes.

- **Function of NK cells**:

- **Elimination** of virally infected cells.

- **Destruction** of cells infected with intracellular bacteria.

- **Destruction** of malignant cells.

- **Rejection** of tissue graft.

- **Release of IFN-gamma** to activate phagocytic cells.

- **Organs of the Immune System**

- Organs of the immune system **are divided into primary and secondary organs:**

- **1) Primary lymphoid organs:**

- **They are the sites for development and maturation of lymphocytes**, in humans they are the thymus and bone marrow.

- **Thymus gland:**

The thymus develops **from the 3rd and 4th pharyngeal pouches** in the **six week of fetal life**.

Located in the upper part of the chest behind **the breast bone (sternum)** and composed of cells similar to lymphocytes.

- **2. Secondary lymphoid organs:**

- They are not essential for generation of lymphocytes but have a **key role in maturation** of these cells and the development of immunity i.e. they organize the immune system.

- **They are:**

- **Lymph nodes.** - **Spleen**

- Lymph nodes surrounding the upper and lower respiratory tracts being known as mucosa-associated lymphoid tissue (MALT).

- Naso pharyngeal lymph nodes,
- Tonsillar lymph nodes,
- Bronchial lymph nodes,

Lymphocytes

Number:

1000 - 1200 in total number **(represent 25-35% of total WBC)** of which 0.1% are renewed daily.

Weight:

almost half as much of the liver.

Size:

6-8 microns in diameter (NK is 10-15 micron).

Shape:

Round (spherical), with darkly stained nucleus fill the whole of cell. **The cytoplasm is scanty**.

- **Types:**

- **T-Lymphocytes 80-90%.**
- **B-Lymphocytes 5-15%.**
- **NK celles 5-10%.**

- **Life Span:**

- **T-cells about 5-10 years** (there is long lived population of T-cells called memory cells).

- **B-cells about 4 days** if they don't encounter antigen, but activated

- **B-cells survive for few weeks** (average 5-7 weeks) with long lived memory cell over 40 years.

- **1. T helper lymphocytes TH: (CD4 cells)**
- **Help the function of other cells** e.g. macrophages, B cells, other helper cells and cytotoxic cells.

- T helper lymphocytes are not a homogenous group of cells but can be divided into subgroups **identified by, cytokines they secrete** and thus the type of help they provide (this subdivision is based on cytokine profiles only, at present time no surface markers or CD numbers are present to distinguish the two subsets).

- **The cytokine profiles subdivide CD4 cells into TH1 and TH2 subsets**

* **TH1 release INF (gamma), IL-2 & IL-12**
* **TH2 release IL 2, 4, 5, 6 & 10**

- **N.B.**: some CD4 cells fall into the two categories since they have a mixed cytokine secretion pattern and are termed T helper 0. (TH0)

- **TH1 tends to switch TH2 and vice versa.**

- **Function of T helper cells (CD4):**

- T helper cells provide inducer (helper) function in T-T, T-B and T-macrophage interactions.

1 - T helper enable cytotoxic cells to respond to antigen to activate suppressor T-cells.

2 - T helper regulates antibody production by B-lymphocyte.

3 - T helper release lymphokines which activate and stimulate macrophages.

- **2. T cytotoxic lymphocyte: (Tc)**

- **"Kill target cells"** TC cells are found predominantly among **CD8** population of cells.

– **MHC unrestricted cytotoxic cells include:**

- NK cells.

- LAK (lymphokine – activated killer cells).

– **MHC restricted cytotoxic cells include:**

- CD8 (Tc) 90% and some CD4 (TH1) 10%.

- 3. **T suppressor lymphocytes: (Ts)**:

- "They down-regulate the immune system":

- In any physiological system there is a key role for **negative** regulation of the action of cells involved and of the immune system i.e.: suppression. e.g.:-

- **Suppression of B-cells and hence antibody production.**

- **Control of cytotoxicity.**

- **Down-regulation of macrophage function.**

- **Function of T lymphocytes:**

- T-lymphocytes are primary involved in:

1. **Cell mediated immunity.**
2. **Delayed** hypersensitivity reaction.
3. **Rejection** of tissue graft.
4. **Elimination** of tumor cells.
5. **Protection against** pathogens which grow intracellularly e.g. virus, fungi, bacteria (T.B. and leprosy).
6. **Regulate antibody** production by B-lymphocytes.
7. **Secrete lymphokines** which affect other cells and regulate the immune response.

Cytokines

- **Definition** :
- Are **soluble glycoproteins** released by cells (cyto) to influence the function (kines) of other cells.

- **Monokines**:
- Are cytokines that are produced by monocytes – macrophages.

- **Lymphokines**:
- Are cytokines produced by lymphocytes.

- **Immunetherapy:**
- GM-CSF is used extensively in patients whose white blood cells have been ablated temporarily as a part of another treatment

  (e.g. antileukemic, cytotoxic therapy).

. Interferons:

First noted as a **part of the innate immune system** as natural antiviral agents.

There are two types of intereferons:-

-Type I , alpha and beta

-Type II , gamma

- **A. <u>Type I Interferon</u> (alpha) and (beta)**

  INF- (Alpha) is produced by : **Monocytes**

  INF- (beta) is produced by : **Fibroblasts**

  Type I IFN is secreted by the virally infected cell into the extra-cellular fluid, it exerts a paracrine effect on adjacent cells to **inhibit virus growth through inhibition of replication of viral RNA or DNA.**

- **<u>Conclusion</u>:**

  Type-I IFN enhance the ability of host defense system to eradicate viruses.

. Tumor necrosis factor (TNF)

- **There are two forms of TNF**:

- TNF (alpha) produced by **Macrophages.**
- TNF (beta) produced by **T cells**
  (stimulated B-cell proliferation

"TNF (beta) is known as lymphotoxin".

- **Functions**:

- TNF is a principle mediator in the host inflammatory response to **gram negative bacteria** (like IL-1 secreted also by macrophages the main stimulus for release is the lipopolysaccharide of bacterial cell wall).

- Local release of TNF **causes migration and activation of neutrophils and macrophages to kill bacteria.**

- It has also **antiparasitic activity.**

* **Systemic release of TNF has the same properties as IL-1. i.e.:**

-- Fever inducing.

-- Production of acute phase proteins .

- The name TNF is derived from early experimental work that it lyses (cytotoxic) tumor cells.

B-Lymphocytes

- **Humoral Immunity**
- B-lymphocyte obtains its name from early studies on antibody production in **birds**, showing that removal of a lymphoid organ known as **bursa of fabricius** from near the hind gut of a chick results in complete inability to produce antibodies.

- These antibody-producing cells then became known as **bursa-derived or B-lymphocytes**.

Immunoglobulin (Antibodies)

- **Definitions:**
- ***Immunoglobulin***

  It is an extremely diverse group of membrane bound or **secreted glycoproteins produced only by B lymphoid cells.**

- ***Antibody***

  It is secreted immunoglobulin protein that is capable of binding specifically to an given antigen specially that cause its production.

- **Nature**:

  Glycoprotein's present in serum and body fluids.

- **Structure of Immunoglobulins**
- 1 = Heavy chain.
- 2 = Light chain
- 3 = disulphide bridge
- 4 = Fc fragment
- 5 = Fab fragment
- 6 = Hinge region

- **Life span of B-lymphocytes**:
- **Is about 4 days** (if they don't encounter antigen),

- But activated B-cells survive of few weeks **(average 5-7 weeks),**

- With long lived memory cells over 40 years **(life span of T cell 5-10 years).**

- **Biological activities of immunoglobulins**

1. Antibodies function on its own:

1) Neutralization when antibodies:

A. Neutralize toxins (antitoxin).

B. Prevent bacterial adherence (anti-bacterial and hence prevent invasion).

C. Prevent virus binding and invading host cell (antiviral).

2) Precipitation when antibodies:

Interact with soluble antigen forming insoluble precipitate.

3) Agglutination:

It is the clumping together of antigen – bearing particles, typically bacteria, RBC or artificially coated latex particles. e.g. agglutinins in autoimmune-hemolytic anemia are usually IgM and directed against RBC antigens.

Agglutinins combine with their targets when the circulation is exposed to slightly lower temperature as in fingers, ear lobes and noses leading to activation of the classical complement pathway.

N.B.: Both precipitation and agglutinationneed antibody with 2 or more antigen binding sites to link with antigen molecules.

- **2. Antibodies function in conjunction with complements:**

When antibodies bind to the surface of the pathogen lead to complement fixation and cell lysis.

- **3. Antibodies function in conjunction with cells:**

1.Opsonization

(is to make bacteria tasty for phagocytosis)

When antibodies coat the antigen enhancing its phagocytosis **opsonizing antibody** through binding between phagocytic cell and its antigen.

2.Antibody dependent cell mediated cytotoxicity (ADCC)

An effector cell kills antibody coated target cells. Killer cells, which able to kill micro-organisms, parasites and infected cells, through Fc receptors recognize and bind to the antibody.

The antibody coat the foreign or infected cell.

The destruction of the target cell occur by releasing toxic molecules

(the process is known as ADCC).

Types of immunoglobulins

Five Classes of immunoglobulins are found dependent on the type of genes encoding the heavy chains present, they are: **Ig G, M, A, D, E.**

IgG

Percentage: 75% of the total serum Ig.

Subclassess:

- IgG1 (60-70%),
- IgG2 (20%),
- IgG3 (8%),
- IgG4 (6%).

- **Antigen binding sites (valence)**: Two

Distribution:

Since it has small molecular weight, it can diffuse to the interstitial fluids.

So, **it is distributed equally between serum and tissue fluids.** IgG (specially IgG1, 3, 4) is the only IgG2 that can **pass the placental barrier providing the new born infant** with humoral antibodies of the mother (**maternal antibodies** or maternal immunity responsible for protection of new born during the first few moths of life).

- **Biological activities**:

  a - Neutralization bacterial toxins.

  b - Fix complement, promoting bacteriolysis (through classical pathway).

  c - Opsonization, enhancing phagocytosis.

- **Stage infections**:
- **IgG is the major antibody in secondary immune response** (predominate late in infections).

IgM:

- **Percentage**: 10% of the total antibodies.
- **Subclassess**: IgM1 and IgM2.
- **Antigen binding sites (valence)**: Ten
- **Distribution**:

  Since it has **large molecules weight**, five time as IgG, (occur primarily as a pentamer composed of 5 IgM monomer joined by J-chain) it tends to remain in the blood vessel **(found only is serum)** and cannot pass through the placental barrier.

  It is an anti A and anti B blood group antibody.

- **Biological activities**:

  The most efficient activator of the complement system (complement is fixed primary with igM and secondary by IgG) **promoting bacteriolysis and phagocytosis.**

- **Stage of infection**:

  IgM is predominant in primary immune response (predominate early in infections) i.e. in response to antigen stimulation IgM predominate early (needs strong antigenic challenge) while in later stages of infection IgG predominates and IgM becomes negligible.

IgA:

- **Percentage**: 15% of total antibodies.

- **Antigen binding sites (valence)**: Four (sIgA)

- **Distribution**:

  IgA is found in serum (monomeric) **and in secretion** (dimeric) e.g. saliva, milk, tears, respiratory secretion, G.I.T. secretion and bronchial secretion.

  Abundant in **"colostrums"** provides immunity for new born infant.

- **Colostrum is yellow fluid secreted by woman's breast few days after new born children**. It contains 20% proteins, sIgA, high amount of salts but less fats and carbohydrates than normal milk.

- **Dmeric** during synthesis of IgA by plasma cells, pair of monomeric IgA are joined with a polypeptide link or "J" chain. Then acquire an additional glycoprotein "secretory piece" to become sIgA.

- **Secretory piece** is an additional polypeptide molecule termed the polyimmunoglobulin receptor (poly IgR) synthesized by epithelial cells lining mucosal surfaces.

- **Biological activities**:

A. Activation of **the complement system** by the alternative pathway.

B. **It coat viruses and bacteria** so that it prevent their attachment and penetration to mucosal surfaces.

C. The sIgA **inhibits colonization of bacteria** on mucosa and teeth.

D. **IgA is an effective opsonin** on monocyte and neutrophils.

IgE (Reaginic antibody)

- **Percentage**: 0.004%
- **Distribution**:

Found in **serum and on the surface of mast cells and basophils.**

It is present in serum of healthy individuals at extremely low levels but found in high levels in atopic patient and in response to **parasitic infections** (e.g. worms).

IgE bind and activate mast cells. Mast cells bear 105 – 106 high affinity IgE receptors.

The interaction between IgE and specific antigen causes mast cell activation and a potent localized and occasionally generalized vascular effects.

It mediates release of histamine and other mediators playing a role in allergic reaction.

IgD

- **Percentage**: 0.004%
- **Distribution**: Found in very **low concentration in serum.**
- **Function**:

The exact function in unknown. It is found binding to receptor sites on the surface of immature B-lymphocytes so it may act for the control of lymphocyte activation and suppression i.e. it may initiate immune response.

Present on the surface of B-cells during maturation so it is:

Involved in maturation of B-cells.

Act as a cell marker for virgin B-cells.

Complement system

Complement was discovered in 1893 by the german scientist "Buchner" and at the beginning of the 20th century by "Charles Border", shortly after the discovery of antibodies.

They demonstrated that if fresh serum containing antibodies was added to the bacteria at physiologic temperature (37oC), the bacteria were lysed.

If serum was heated to 56^{o}C or more, it lost its ability to kill bacteria. This was not due to destruction of antibody activity, because antibodies are heat stable.

- **Pathways of complement activation**
- **There are two main distinct parallel** but entirely independent pathways for activation of the complement system, **the classical and the alternative (properdin) pathway.**

- **1. The classical pathway:**
- Stats by activation of C1, C4, C2, C3, → C9.

- **2. The alternative pathway**
  (properdin pathway)
- Although the alternative pathway was discovered second, it is the oldest of the pathways.

- **Biological effects of complement activation**
- Activation of the complement system can mediate different biological effects.

- It plays a major role in host defence through immune regulation, mediation of inflammation, immune adherence, chemotaxis, opsonization, phagocytosis, clearance of the immune complex, increase of the bactericidal activity of phagocytes and cell lysis.

- It should be remembered that the above mentioned activities are protective yet the complement system activation may contribute to unwanted damage and tissue injury as a result of excessive inflammatory response.

- **1. <u>Complement – mediated immune regulation</u>:**
- **C3a, C5a and C3b** complement fragments are immuno-regulatory as:
  - C3a decreases T cell function.
  - C5a increases T cell function.
  - C5a stimulates IL-1 and IL-6 production by monocytes and macrophages. While IL-1 stimulates T-helper cells, IL-6 promotes growth and differentiation of B-cells.
  - C3a and C5a enhanced antibody production.
- **C3b regulates immunological memory of B-lymphocyte**

- **2. <u>Complement-mediated inflammation</u>:**
- C3a, C4a and C5a complement fragments are **known as anaphylatoxins.**

- This name is derived from their putative role in the clinical syndrome anaphylaxis.

- **Anaphylatoxins** can attach to their corresponding receptors (anaphylatoxin receptors) **present on mast cells and basophils** causing non-cytolytic **release of histamine and other mediators of inflammation.**

- While **C5a is the most potent anaphylatoxin** C4a is the least.

- **Mast cell and basophil granules** are categorized as preformed interacellular granules and substances synthesized from the mast cell wall upon activation.

- **I. Preformed granules:**

1. Histamine:

- **Vasodilatation** of the blood vessels and increased capillary permeability.
- **Contraction** of smooth muscles.
- Stimulation of glandular secretion.

2. Heparin:

- Anticoagulant.
- Activate C1q.

- ***3. Complement-mediated chemotaxis:***

Chemotaxis is the attraction and directional movement of phagocytic cells to the areas of increased concentration of attracting molecules (termed chemo-attractant).

C5a, in a very low concentration, is the most potent chemotatactic factor derived from the complement system.

- **C5a** and **C3a** are chemotatactic for neutrophils, eosinophils and macrophages.
- **C5b67** is chemotatactic for neutrophils and eosinophils.

- ***4. Complement-mediated opsonization***:

  Opsonization is to stimulate phagocytosis by antibody and/or complement fragment that render the particle more attractive for phagocytic cells.

  C3b accounts for most of the complement opsonic activity and foreign particles are phagocytosed better if coated with C3b.

  Foreign particles coated with C3b bind to C3b receptors (CR-1) on surface of phagocytes **facilitating phagocytosis** i.e. complement fragment C3b serves as a bridge between foreign particles and phagocytes.

- ***5. Complement-mediated cell lysis and death:***

  C567 binds and insert itself into the cell membrane forming transmembrane unit (TMU) **C5b678(9)** is the membrane attack complex (MAC). Holes and transmembrane channels are made in the cell membrane through which small ions and water can pass to the cell leading to cell rupture, destruction and death through osmotic lysis.

THANK YOU